Chapter 4: Consumer and Producer Surplus, Market Efficiency, and Welfare

Principles of Microeconomics Curriculum

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Abstract

This chapter provides a rigorous, pedagogical treatment of Chapter 4: Consumer and Producer Surplus, Market Efficiency, and Welfare, benchmarked to standard principles curriculum.

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1 Surplus Metrics

Consumer Surplus (CS) is the area under inverse demand above market price:

$$CS = \int_0^{Q^*} [P(q) - P^*]\, dq = \frac{1}{2} Q^* (P_{\text{choke}} - P^*) \tag{1}$$

Producer Surplus (PS) is the area above inverse supply below market price:

$$PS = \int_0^{Q^*} [P^* - MC(q)]\, dq = \frac{1}{2} Q^* (P^* - P_{\min}) \tag{2}$$

Total Surplus ($TS = CS + PS$) is maximized at competitive market equilibrium where $MSB = MSC$.

2 Deadweight Loss of Taxation

An excise tax t creates a tax wedge $P_b - P_s = t$. Deadweight loss follows Harberger's formulation:

$$DWL = \frac{1}{2} \times t \times (Q^* - Q_t) \tag{3}$$

The relative tax incidence ratio depends on elasticities: $\frac{\Delta P_b}{\Delta P_s} = \frac{\varepsilon_s}{|\varepsilon_d|}$.